

PHYSICIAN & DEPARTMENT SQL

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The screenshot shows the SQL Developer interface with the following SQL script in the editor:

```
1  -- Step 1: Create Database and Physician Table
2  • CREATE DATABASE physician_department_db;
3  • USE physician_department_db;
4
5  • CREATE TABLE physician (
6      employeeid INT PRIMARY KEY,
7      name VARCHAR(100),
8      position VARCHAR(100),
9      ssn VARCHAR(20)
10 );
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
✓ 1	13:44:04	CREATE DATABASE physician_department_db	1 row(s) affected	0.016 sec
✓ 2	13:44:04	USE physician_department_db	0 row(s) affected	0.000 sec
✓ 3	13:44:04	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VAR...	0 row(s) affected	0.031 sec

The screenshot shows the SQL Developer interface with the following SQL script in the editor:

```
10 )
11 -- Step 2: Insert Data into Physician
12 • INSERT INTO physician VALUES
13 (1, 'Hima', 'Staff Internist', '111111111'),
14 (2, 'Ravi', 'Attending Physician', '222222222'),
15 (3, 'Siri', 'Surgical Attending Physician', '333333333'),
16 (4, 'Santhi', 'Senior Attending Physician', '444444444'),
17 (5, 'Varshika', 'Head Chief of Medicine', '555555555'),
18 (6, 'Manu', 'Surgical Attending Physician', '666666666'),
19 (7, 'Shiny', 'Surgical Attending Physician', '777777777'),
20 (8, 'Pranitha', 'MD Resident', '888888888'),
21 (9, 'Angela', 'Attending Psychiatrist', '999999999');
22
```

The Output window shows the execution results:

#	Time	Action	Message	Duration / Fetch
✓ 1	13:44:04	CREATE DATABASE physician_department_db	1 row(s) affected	0.016 sec
✓ 2	13:44:04	USE physician_department_db	0 row(s) affected	0.000 sec
✓ 3	13:44:04	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VAR...	0 row(s) affected	0.031 sec
✓ 4	13:45:58	INSERT INTO physician VALUES (1, 'Hima', 'Staff Internist', '111111111'). (2, '...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0.032 sec

```

16 (4, 'Santhi', 'Senior Attending Physician', '44444444'),
17 (5, 'Varshika', 'Head Chief of Medicine', '55555555'),
18 (6, 'Manu', 'Surgical Attending Physician', '66666666'),
19 (7, 'Shiny', 'Surgical Attending Physician', '77777777'),
20 (8, 'Pranitha', 'MD Resident', '88888888'),
21 (9, 'Angela', 'Attending Psychiatrist', '99999999');
22
23 -- Step 3: Create Department Table
24 CREATE TABLE department (
25     departmentid INT PRIMARY KEY,
26     name VARCHAR(100),
27     head INT
28 );

```

Automatic context help is disabled. Use the toolbar icon to manually get help for the current caret position or to toggle automatic help.

Output

Action Output

#	Time	Action	Message
1	13:44:04	CREATE DATABASE physician_department_db	1 row(s) affected
2	13:44:04	USE physician_department_db	0 row(s) affected
3	13:44:04	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), head INT);	0 row(s) affected
4	13:45:58	INSERT INTO physician VALUES (1, 'Hima', 'Staff Internist', '1111111111'), (2, 'Hima', 'Staff Internist', '1111111111'), (3, 'Hima', 'Staff Internist', '1111111111'), (4, 'Hima', 'Staff Internist', '1111111111'), (5, 'Hima', 'Staff Internist', '1111111111'), (6, 'Hima', 'Staff Internist', '1111111111'), (7, 'Hima', 'Staff Internist', '1111111111'), (8, 'Hima', 'Staff Internist', '1111111111'), (9, 'Hima', 'Staff Internist', '1111111111');	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0
5	13:46:54	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT);	0 row(s) affected

```

36 -- Step 5: INNER JOIN
37 SELECT d.name AS Department,
38        p.name AS Head_Physician
39 FROM department d
40 JOIN physician p
41 ON d.head = p.employeeid;

```

Automatic context help is disabled. Use the toolbar icon to manually get help for the current caret position or to toggle automatic help.

Result Grid

Department	Head_Physician
General Medicine	Santhi
Surgery	Shiny
Psychiatry	Angela

Result 1

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	13:44:04	USE physician_department_db	0 row(s) affected	0.000 sec
3	13:44:04	CREATE TABLE physician (employeeid INT PRIMARY KEY, name VARCHAR(100), head INT);	0 row(s) affected	0.031 sec
4	13:45:58	INSERT INTO physician VALUES (1, 'Hima', 'Staff Internist', '1111111111'), (2, 'Hima', 'Staff Internist', '1111111111'), (3, 'Hima', 'Staff Internist', '1111111111'), (4, 'Hima', 'Staff Internist', '1111111111'), (5, 'Hima', 'Staff Internist', '1111111111'), (6, 'Hima', 'Staff Internist', '1111111111'), (7, 'Hima', 'Staff Internist', '1111111111'), (8, 'Hima', 'Staff Internist', '1111111111'), (9, 'Hima', 'Staff Internist', '1111111111');	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0.032 sec
5	13:46:54	CREATE TABLE department (departmentid INT PRIMARY KEY, name VARCHAR(100), head INT);	0 row(s) affected	0.031 sec
6	13:47:41	INSERT INTO department VALUES (1, 'General Medicine', 4), (2, 'Surgery', 7), (3, 'Psychiatry', 9);	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
7	13:47:41	SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.head = p.employeeid;	3 row(s) returned	0.000 sec / 0.000 sec

Physician & Department SQL

```

29 (1,'General Medicine',4),
30 (2,'Surgery',7),
31 (3,'Psychiatry',9);
32 -- Step 5: INNER JOIN
33 • SELECT d.name AS Department,
34       p.name AS Head_Physician
35 FROM department d
36 JOIN physician p
37 ON d.head = p.employeeid;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Department	Head_Physician
General Medicine	Santhvin
Surgery	Shiny
Psychiatry	Angela

```

47 -- Step 7: WHERE + LIKE
48 • SELECT *
49 FROM physician
50 WHERE name LIKE 'John%';
51

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

employeeid	name	position	ssn
*	NULL	NULL	NULL

physician 4 x | Apply | Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	13:46:54	CREATE TABLE department (departmentid INT PRIMARY KEY, n...	0 row(s) affected	0.031 sec
6	13:47:41	INSERT INTO department VALUES (1,'General Medicine', 4), (2,'Surge...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.031 sec
7	13:47:41	SELECT d.name AS Department, p.name AS Head_Physician FRO...	3 row(s) returned	0.000 sec / 0.000
8	13:49:12	SELECT * FROM physician WHERE position = 'Surgical Attending Physi...	3 row(s) returned	0.015 sec / 0.000
9	13:49:12	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000
10	13:49:16	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000

Automatic context help disabled. Use the tool manually get help for current caret position toggle automatic help

BankDB* SQL File 2* x

```

58
59 -- Step 10: GROUP BY
60 • SELECT position,
61       COUNT(*) AS Total_Employees
62 FROM physician
63 GROUP BY position;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

position	Total_Employees
Staff Internist	1
Attending Physician	1
Surgical Attending Physician	3
Senior Attending Physician	1
Head Chief of Medicine	1

Result 5 | Result 6 | Result 7 x | Read Only | Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration /
8	13:49:12	SELECT * FROM physician WHERE position = 'Surgical Attending Physi...	3 row(s) returned	0.015 sec.
9	13:49:12	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned	0.000 sec.
10	13:49:16	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned	0.000 sec.
11	13:49:50	SELECT COUNT(*)AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned	0.015 sec.
12	13:49:50	SELECT COUNT(DISTINCT position)AS Different_Positions FROM phy...	1 row(s) returned	0.000 sec.
13	13:49:50	SELECT position, COUNT(*)AS Total_Employees FROM physician ...	7 row(s) returned	0.000 sec.

Automatic conte disabled. Use the manually get he current caret pos toggle automa

70
71 -- Step 12: ORDER BY ASC
72 • SELECT *
73 FROM physician
74 ORDER BY name ASC;
75

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

employeeid	name	position	ssn
9	Angela	Attending Psychiatrist	999999999
1	Hima	Staff Internist	111111111
6	Manu	Surgical Attending Physician	666666666
8	Pranitha	MD Resident	888888888
2	Ravi	Attending Physician	222222222

Result 8 physician 9

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 10	13:49:16	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.0
✓ 11	13:49:50	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned	0.015 sec / 0.0
✓ 12	13:49:50	SELECT COUNT(DISTINCT position) AS Different_Positions FROM phy...	1 row(s) returned	0.000 sec / 0.0
✓ 13	13:49:50	SELECT position, COUNT(*) AS Total_Employees FROM physician ...	7 row(s) returned	0.000 sec / 0.0
✓ 14	13:50:43	SELECT position, COUNT(*) AS Total_Employees FROM physician ...	1 row(s) returned	0.000 sec / 0.0
✓ 15	13:50:43	SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.0

70 • SELECT *
71 FROM physician
72 WHERE employeeid IN
73 (
74 SELECT head
75 FROM department
76)

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

Result Grid

employeeid	name	position	ssn
4	Santhi	Senior Attending Physician	444444444
7	Shiny	Surgical Attending Physician	777777777
9	Angela	Attending Psychiatrist	999999999
...	...	...	...

physician 10 Result 11 Result 12 Result 13 physician 14 physician 15

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 16	13:51:34	SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
✓ 17	13:51:34	SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
✓ 18	13:51:34	SELECT COUNT(DISTINCT position) AS Different_Positions FROM phy...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 19	13:51:34	SELECT position, COUNT(*) AS Total_Employees FROM physician ...	7 row(s) returned	0.000 sec / 0.000 sec
✓ 20	13:51:34	SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000	9 row(s) returned	0.000 sec / 0.000 sec
✓ 21	13:51:34	SELECT * FROM physician WHERE employeeid IN (SELECT head ...	3 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

101 FROM physician;
102
103 -- Step 24: MIN()
104 • SELECT MIN(employeeid) AS Lowest_Employee_ID
105 FROM physician;
106
107

```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Result Grid

	employeeid	name	position	ssn
1	Hima	Staff Internist	111111111	
2	Ravi	Attending Physician	222222222	
4	Santhi	Senior Attending Physician	444444444	
5	Varshika	Head Chief of Medicine	555555555	
8	Pranitha	MD Resident	888888888	

physician 16 × Result 17 department 18 Result 19 Result 20 Apply Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	13:51:34	SELECT * FROM physician WHERE employeeid IN (SELECT head ...	3 row(s) returned	0.000 sec / 0.000 sec
22	13:52:39	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LI...	6 row(s) returned	0.016 sec / 0.000 sec
23	13:52:39	SELECT position, COUNT(*) AS Total FROM physician GROUP BY ...	1 row(s) returned	0.016 sec / 0.000 sec
24	13:52:39	SELECT * FROM department d WHERE EXISTS (SELECT 1 FRO...	3 row(s) returned	0.000 sec / 0.000 sec
25	13:52:39	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician L...	1 row(s) returned	0.000 sec / 0.000 sec
26	13:52:39	SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physician LI...	1 row(s) returned	0.000 sec / 0.000 sec

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

101 FROM physician;
102
103 -- Step 24: MIN()
104 • SELECT MIN(employeeid) AS Lowest_Employee_ID
105 FROM physician;
106
107

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Result Grid

	Highest_Employee_ID
1	9

physician 16 Result 17 department 18 Result 19 × Result 20 Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	13:51:34	SELECT * FROM physician WHERE employeeid IN (SELECT head ...	3 row(s) returned	0.000 sec / 0.000 sec
22	13:52:39	SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LI...	6 row(s) returned	0.016 sec / 0.000 sec
23	13:52:39	SELECT position, COUNT(*) AS Total FROM physician GROUP BY ...	1 row(s) returned	0.016 sec / 0.000 sec
24	13:52:39	SELECT * FROM department d WHERE EXISTS (SELECT 1 FRO...	3 row(s) returned	0.000 sec / 0.000 sec
25	13:52:39	SELECT MAX(employeeid) AS Highest_Employee_ID FROM physician L...	1 row(s) returned	0.000 sec / 0.000 sec
26	13:52:39	SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physician LI...	1 row(s) returned	0.000 sec / 0.000 sec